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Urine bag holding bracket

Abstract

A portable urine collection bag holding bracket is an easy to use and manufacture, low cost and quickly detachable and fixable bracket that facilitates attachment of a urine collection bag or the like to a patient support system or other support at a desired adjustable position based on the user's requirement. The portable urine collection bag holding bracket comprises an L-shaped bracket, wherein the corner area of the L-shaped bracket comprises magnets that allow quick attachment and detachment of the L-shaped bracket to the bed, seat, wheelchair, stretcher, IV stand, or other patient support system or other support. Further, the L-shaped bracket is provided with a plurality of bracket holes to accommodate various configurations of hooks or hangers of the urine collection bag(s) and facilitate easy and quick attachment and detachment of the urine collection bag(s) with respect to the L-shaped bracket.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. provisional application No. 63/323,206, filed Mar. 24, 2022, and entitled URINE BAG HOLDING BRACKET, which provisional application is incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates generally to a bag holder. More so, the present invention relates to a portable urine bag holder.

BACKGROUND OF THE INVENTION

(2) Generally, urine bags are used for patients in hospitals, nursing homes and other facilities when required such as, for example, after major surgical operations. After surgery, the surgical patient may be transferred from the operating room to an intensive care unit or general ward which is equipped with catheters and urine collection bags. A urine collection bag is typically connected to a catheter, which is inserted in the urethra of the patient. The urine collection bag may be supported by an infusion stand adjacent to the patient's bed or wheelchair, for example.

(3) Numerous attempts have been made and several prior art devices are known for variety of holders. Even though these innovations may be suitable for the specific purposes to which they address, however, they would not be as suitable for the purposes of the present invention.

(4) For example, patent number TWM433852U to Peng Qizong et al. discloses a urine bag holder which is a hanger configured to hang a urine bag on a support.

(5) Patent number GB8817131D0 to Mcleod teaches a hanger for suspending an article such as a urine drainage bag which has spaced-apart holes to hang the bag.

(6) Patent number DE9400135U1 to Dr Med discloses a complex support device for secretion bags. The support device has a hook-shaped upper part and a support plate connected thereto with a support for a secretion bag. The design is suitable for attachment to a patient support device but does not fit to any type of horizontal surface.

(7) Patent number DE8235580U1 describes a secretion bag holder comprising a clamping plate with a hinge connected to a holding plate to hold a secretion bag.

(8) Patent number CN215131300U to Cao Lei, et al. discloses a urine bag holding device comprising a waist belt for patient. A slidable locking mechanism on the belt can move along the belt while holding the urine bag.

(9) Patent number CN215023324U to Yang et al. discloses a urine bag hang-ng fixture comprising an L-shaped mount that is provided with a measuring mechanism and holds the urine bag in position. However, the design is complex, expensive, and difficult to manufacture.

(10) Patent number CN214970875U to Cai Xiaoli, et al. teaches an adjustable catheter fixing mechanism comprising a waist belt for a patient. One or more hangers is provided on the waist belt. A catheter holder on the waist belt is configured to adjustably hold a catheter and urine bag in place on the belt.

(11) There are also other conventional ways to support a urine bag on a patient support system, such as by applying sticky tape, using cable ties, Velcro, rope, or the like to detachably hold the urine bag to the system. But all the above examples have some or the other demerits such as being expensive or difficult to manufacture, having a complex design or being bulky or inefficient. Thus, there is a need of the present invention to provide a portable urine bag holder that addresses and provides a straightforward design which is easy to use and manufacture, is low-cost and can conveniently hold a urine bag or the like at the desired position with provision to quickly change and fix the holder.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The invention will now be described, by way of example, with reference to the accompanying drawings, in which:
- (2) FIG. 1 illustrates a perspective view of an exemplary portable urine bag holding bracket, in accordance with an illustrative embodiment of the present invention;
- (3) FIG. 2 illustrates a top view of the illustrative portable urine bag holding bracket;
- (4) FIG. 3 illustrates a front view of the illustrative portable urine bag holding bracket; and
- (5) FIG. 4 illustrates a front view of a slot of the illustrative portable urine bag holding bracket.
- (6) Like reference numerals refer to like parts throughout the various views of the drawings.

DETAILED DESCRIPTION OF THE INVENTION

(7) The following detailed description is merely exemplary in nature and is not intended to limit the described embodiments or the application and uses of the described embodiments. As used herein, the word “exemplary” or “illustrative” means “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” or “illustrative” is not necessarily to be construed as preferred or advantageous over other implementations. All of the implementations described below are exemplary implementations provided to enable persons skilled in the art to make or use the embodiments of the disclosure and are not intended to limit the scope of the disclosure, which is defined by the claims. For purposes of description herein, the terms “upper,” “lower,” “left,” “rear,” “right,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the invention as oriented in FIG. 1. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the inventive concepts defined in the appended claims. Specific dimensions and other physical characteristics relating to the embodiments disclosed herein are therefore not to be considered as limiting unless the claims expressly state otherwise.

(8) In typical use of a urine collection bag attached to a catheter which is inserted in a patient's urethra, as soon as urine leaves the patient's bladder and enters the catheter, bacteria begin to accumulate in the catheter. If it is not properly drained from the bladder, the urine instantly reenters the patient's bladder. This places the patient at risk for a Urinary Tract Infection (UTI) which can spread upward into the kidneys and cause serious complications to the patient, including kidney damage or sepsis. Currently, the catheter and the urine collection bag may be placed on the floor or suspended from any nearby object which may be so high that proper drainage of urine into the urine collection bag is restricted. If the patient is subsequently transferred, the urine collection bag may be tossed onto the patient and then suspended on a stationary component of the bed that does not facilitate proper drainage of the urine from the patient. This may be a significant problem at nursing facilities or home or during travel. The portable urine bag holding bracket **100** of the present invention, when used in conjunction with evidence-based practice, can assist in preventing the above situations and facilitate ease in drainage of urine from the patient.

(9) According to some embodiments of the present invention, a portable urine bag holding bracket, hereinafter bracket **100**, as shown in FIGS. 1-4 is an easy-to-use and manufacture, low-cost and quickly detachable and fixed bracket that allows a urine collection bag or the like to be supported at a desired adjustable position based on the user's requirements. The bracket **100** may include a L-shaped bracket. The corner area of the L-shaped bracket **100** may be magnetized to facilitate quick attachment and detachment of the bracket **100** to a bed, seat, or wheelchair, stretcher, IV stand, or similar patient support system. Further, as will be hereinafter described, the bracket **100** may be provided with a plurality of bracket holes **103** to accommodate hooks or hangers for the urine

collection bag or bags in various numbers and configurations or orientations.

(10) According to one aspect of the present invention as shown in FIG. 1, a portable urine collection bag holding bracket **100** may include a first bracket portion **102**. The first bracket portion **102** may include a first proximal segment **106** and a first distal segment **108**. The first proximal segment **106** may be elongated. The first distal segment **108** of the first bracket portion **102** may be magnetized. The first proximal segment **106** may include a plurality of bracket holes **103**.

(11) The urine bag holding bracket **100** may further include a second bracket portion **104**. The second bracket portion **104** may include a second proximal segment **110** and a second distal segment **112**. The second proximal segment **110** may be elongated. The second distal segment **112** of the second bracket portion **104** may be magnetized. The second proximal segment **110** may include a plurality of bracket holes **103**. Further, the first distal segment **108** of the first bracket portion **102** and the second distal segment **112** of the second bracket portion **104** may meet orthogonally to form the bracket **100**.

(12) According to another aspect of the present invention, the bracket **100** may be “L-shaped”.

(13) According to another aspect of the present invention, the first bracket portion **102** of the bracket **100** may be 8-16 inch long.

(14) According to another aspect of the present invention, the second bracket portion **104** of the bracket **100** may be 6-10 inch long.

(15) According to another aspect of the present invention, the first bracket portion **102** and the second bracket portion **104** of the bracket **100** may be 1-2 inch wide and 1-5 mm thick.

(16) According to another aspect of the present invention, the first bracket portion **102** may comprise 4-7 bracket holes **103**.

(17) According to another aspect of the present invention, the first bracket portion **102** may comprise 3-5 bracket holes **103**.

(18) According to another aspect of the present invention, the bracket holes **103** may be disposed in aligned and spaced-apart relationship to each other along the longitudinal axis of the corresponding first proximal segment **106** of the first bracket portion **102** and second proximal segment **110** of the second bracket portion **104**.

(19) According to another aspect of the present invention, each of the bracket holes **103** may resemble a human head configuration to allow quick and adjustable attachment and detachment of hooks or hangers for the urine collection bag or bags in various numbers and configurations.

(20) According to another aspect of the present invention, each bracket hole **103** may have a multi-faceted shape with a curved or concave first edge **120**; a pair of straight, parallel, spaced-apart side edges **121** extending from the first edge **120**; a pair of straight outer stepped edges **122** extending inwardly from the respective side edges **121**; a pair of straight sloped edges **123** angling inwardly from the respective outer stepped edges **122**; a pair of straight inner lower stepped edges **124** extending inwardly from the respective sloped edges **123**; and a curved or concave second edge **125** extending between the inner stepped edges **124**.

(21) One objective of the present invention is to provide a portable urine collection bag holding bracket **100** that is easy to use, manufacture and transport.

(22) Another objective of the present invention is to provide a portable urine collection bag holding bracket **100** that is provided with a possible magnetized portion, thereby allowing quick attachment and detachment of the urine collection bag or bags to the bracket **100**.

(23) Yet another objective of the present invention is to provide a portable urine collection bag holding bracket **100** that is provided with multiple bracket holes **103** at predetermined distances and spacings along the first proximal segment **106** of the first bracket portion **102** and along the second proximal segment **110** of the second bracket portion **104**, thereby allowing adjustable attachment of the urine collection bags to the bracket **100** at desired positions or heights as per the requirement.

(24) Another objective of the present invention is to provide a portable urine collection bag holding

bracket **100** that is that is provided with multiple bracket holes **103**, wherein each of the bracket holes **103** may resemble a human head configuration to allow quick and adjustable attachment and detachment of hooks or hangers for the urine collection bags in various numbers and configurations.

(25) According to an illustrative embodiment of the present invention, FIG. **1** illustrates a perspective view of the exemplary portable urine bag holding bracket **100**, wherein the bracket comprises a first bracket portion **102** and a second bracket portion **104**. The first bracket portion **102** of the bracket **100** may be 8-16 inch long, preferably 10 inch long. The second bracket portion **104** of the bracket **100** may be 6-10 inch long, preferably 7 inch long. The first bracket portion **102** and the second bracket portion **104** of the bracket **100** may be approximately 2 inch wide and 1-5 mm thick. Further, the first bracket portion **102** may include 4-7 bracket holes **103**, and preferably, 5 bracket holes **103**. However, the second bracket portion **104** of the bracket **100** may include 3-5 bracket holes **103**, and preferably, 3 bracket holes **103**. The bracket holes **103** may be disposed in aligned and spaced-apart relationship to each other along the longitudinal axis of the corresponding first proximal segment **106** of the first bracket portion **102** and second proximal segment **110** of the second bracket portion **104**.

(26) Each of the bracket holes **103** may resemble a human head configuration, as shown in FIG. **4**. Accordingly, each bracket hole **103** may have a multi-faceted shape with a curved or concave first edge **120**; a pair of straight, parallel, spaced-apart side edges **121** extending from the first edge **120**; a pair of straight outer stepped edges **122** extending inwardly from the respective side edges **121**; a pair of straight sloped edges **123** angling inwardly from the respective outer stepped edges **122**; a pair of straight inner lower stepped edges **124** extending inwardly from the respective sloped edges **123**; and a curved or concave second edge **125** extending between the inner stepped edges **124**.

(27) The first bracket portion **102** and the second bracket portion **104** may meet orthogonally to form the bracket **100** that is “L-shaped”. Further, the first distal segment **108** of the first bracket portion **102** and the second distal segment **112** of the second bracket portion **104** may be magnetized to facilitate quick attachment and detachment of the bracket **100** to the bed, seat, wheelchair, stretcher, IV stand, or other patient support system (not illustrated). The L-shaped bracket **100** may be configured to strongly attach to the edge of the patient support system such that the first bracket portion **102** or the second bracket portion **104** interchangeably hangs downwardly from the patient support system to hold the urine collection bag at the desired position.

(28) FIG. **2** illustrates a top view of the exemplary portable urine bag holding bracket **100**, in accordance with an illustrative embodiment of the present invention. FIG. **3** illustrates a front view of the exemplary portable urine bag holding bracket **100**, in accordance with an illustrative embodiment of the present invention, and illustrates the respective first and second distal segments **108**, **112** that may be magnetized and the respective first and second proximal segments **106**, **110** that are non-magnetized portions of the respective first bracket portion **102** or the second bracket portion **104** of the bracket **100**. The multiple bracket holes **103**, provided at different heights, may facilitate adjustment in the hanging positions of the urine collection bag as per the need of the patient. The straightforward design of the bracket **100** allows easy alignment of the catheter with respect to the urine collection bag while facilitating ease in visualization of the urine output and drainage into the urine collection bag without any blockage, stagnant or reverse flow of urine towards the patient.

(29) The bracket **100** of the present invention can be used for any other applications without limiting from the scope and spirit of the present invention.

(30) According to another exemplary embodiment of the present invention, the bracket **100** can be manufactured using any material including metal or non-metal, wherein the first and second distal segments **108**, **112** of the respective first bracket portion **102** and second bracket portion **104** may be embedded or glued with magnets or magnetized via other technology known in the art for a desired length of 1-2 inch. These first and second distal segments **108**, **112** of the respective first

bracket portion **102** or the second bracket portion **104** of the bracket **100** may meet along an orthogonal meeting line **114** to form the L-shaped bracket **100**.

(31) In typical application, the bracket **100** may support a urine collection bag (not illustrated) on a bed, seat, wheelchair, stretcher, IV stand, or other suitable patient support system or other support (not illustrated) in a position which is optimal to facilitate efficient drainage of urine from the patient through a catheter attached to the urine support bag. Accordingly, the typically magnetic first distal segment **108** of the first bracket portion **102** and/or second distal segment **112** of the second bracket portion **104** may be magnetically attached to a magnetic element on the patient support system or other support. A hook or hanger (not illustrated) on the urine collection bag may be placed into engagement with one of the vertically spaced bracket holes **103** in the first proximal segment **106** of the first bracket portion **102** or with one of the horizontally spaced bracket holes **103** in the second proximal segment **110** of the second bracket portion **104**. The choice of which set of the bracket holes **103** is used for the purpose may depend on such factors as the type of patient support system or other support which is used to support the urine collection bag, as well as the vertical and/or horizontal position of the urine support bag with respect to the patient. The unique shape and configuration of each bracket hole **103** may be particularly well-suited to the particular type of hook or hanger which is commonly used to attach the urine support bag to the bracket **100**.

(32) Subsequent removal of the full urine collection bag may be accomplished by disengagement of the hook or hanger from the bracket hole **103**. A replacement urine collection bag may be attached to the same or a different bracket hole **103**, as needed. The bracket **100** may be removed from the patient support system or other support by detaching the first distal segment **108** of the first bracket portion **102** and/or the second distal segment **112** of second bracket portion **104** from the magnetic element on the system or other support.

(33) These and other advantages of the invention will be further understood and appreciated by those skilled in the art by reference to the following written specification, claims and appended drawings.

(34) Because many modifications, variations, and changes in detail can be made to the described preferred embodiments of the invention, it is intended that all matters in the foregoing description and shown in the accompanying drawings be interpreted as illustrative and not in a limiting sense. Thus, the scope of the invention should be determined by the appended claims and their legal equivalence.

Claims

1. A portable urine collection bag holding bracket, wherein the bracket comprises: a first bracket portion, wherein the first bracket portion comprises an elongated first proximal segment and a first distal segment extending from the first proximal segment, and the first proximal segment comprises a first plurality of aligned, spaced-apart, multi-faceted bracket holes; a second bracket portion, wherein the second bracket portion comprises an elongated second proximal segment and a second distal segment, wherein the second distal segment is magnetized, and the second proximal segment comprises a second plurality of aligned, spaced-apart, multi-faceted bracket holes, further the first distal segment of the first bracket portion and the second distal segment of the second bracket portion meet orthogonally at an orthogonal meeting line to form the bracket; and wherein each of the first plurality of aligned, spaced-apart, multi-faceted bracket holes comprises: a curved or concave first edge; a pair of straight, parallel, spaced-apart side edges extending from the first edge; a pair of straight outer stepped edges extending inwardly from the side edges, respectively; a pair of straight sloped edges angling inwardly from the outer stepped edges, respectively; a pair of straight inner lower stepped edges extending inwardly from the sloped edges, respectively; and a curved or concave second edge extending between the inner lower stepped edges.

2. The bracket of claim 1, wherein the bracket is “L”-shaped.

3. The bracket of claim 1, wherein the first distal segment is magnetized.
 4. The bracket of claim 1, wherein the second bracket portion is 6-10 inch long.
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